Van Nguyen
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RESEARCH INTEREST

Applied Mathematics and Statistics, Biostatistics, Statistical Genomics, Single-Cell Perturbation Analysis, Bioinformatics, Sequence Alignment, Genomics and Proteomics Analysis, AI-Driven Medical Imaging, Optimal Transport, Uncertainty Quantification, Systems Biology, Precision Medicine, Large Language Models in Biomedical Research, Quantitative Finance.

Work and Research Experience

- **Research Collaboration: Single-Cell Perturbation Screens**
  Statistical Methodology Research *Feb 2026–Oct 2026*
  Manuscript: "Closed-form sample size calculation for estimating perturbation directions in single-cell screens" (Biostatistics, Under Review)
  - **Statistical Methodology**: Developed a theoretical framework using the **Multivariate Delta Method, Taylor Expansion, and Asymptotic Variance** to quantify uncertainty in drug-induced perturbation direction estimates.
  - **Sample Size Estimation**: Derived closed-form sample size approximations based on **Covariance Propagation, Signal-to-Noise Ratio, and Directional Estimation Precision**.
  - **Drug-Response Analysis**: Investigated the effects of biological and technical noise on perturbation direction estimation and drug-response signals in single-cell screening experiments.

- **Research Collaboration: CBCT-Guided Adaptive Radiotherapy** **USA**
  CBCT for Rectal Cancer Adaptive Treatment *Mar 2026–Present*
  Collaborators: Dr. Duong Thanh Tai (Spencer Hospital, USA); Dr. Dat Tran (Louisiana State University, USA)
  - **Gas-Aware Deformable Registration**: Formulating CBCT–CT registration as a **diffeomorphic deformation** coupled with a **Wasserstein–Fisher–Rao (unbalanced optimal transport)** source term to model rectal gas and bladder-filling changes, benchmarked against **VoxelMorph and TransMorph** on contour propagation and dose accumulation.
  - **Risk-Controlled Segmentation**: Wrapping **Swin UNETR and Bayesian Deep Ensembles** in **Conformal Risk Control** with **Mondrian and Weighted Conformal Prediction**, producing contour bands with finite-sample guarantees on missed CTV volume to flag cases for radiation oncologist review.
  - **Counterfactual Dosimetry**: Using **Neural Dose Surrogates, Bilevel Optimization, and Differentiable DVH Loss** to quantify the dosimetric impact of residual segmentation errors.
  - **Data and Validation**: Exploring **SynthRAD2025, COBRA2026**, and institutional rectal CBCT/CT datasets with expert contours and radiotherapy dose records for geometric and dosimetric evaluation, subject to data availability.

- **Bioinformatics Research Intern: Entobel** **Singapore**
  Researcher Intern *Jun 2026–Sep 2026*
  - **Research Project**: *Development of a Nextflow Pipeline and Report Template for BSF Genomic Diversity and Inbreeding Monitoring.*
  - **Pipeline Development**: Designed and implemented a scalable **Nextflow-based bioinformatics workflow** for automated processing and analysis of Black Soldier Fly (BSF) genomic sequencing data.
  - **Genomic Diversity and Inbreeding Analysis**: Developed analytical modules for genomic diversity assessment, population structure evaluation, and inbreeding coefficient monitoring across BSF breeding populations.
  - **Workflow Automation and Reporting**: Integrated quality control, variant calling, statistical summarization, and automated report generation into a reproducible computational framework for genomic monitoring.
  - **Industrial Bioinformatics Collaboration**: Collaborated with research and technical teams to optimize computational reproducibility and support data-driven breeding strategy decisions for sustainable industrial insect production.

- **Research Assistant Project: "Causal Narrative in Information Systems Research"** **USA**
  Researcher Collaboration *Jul 2025–Dec 2025*
  Minh Nguyen (Florida Atlantic University) and Mike Nguyen (University of Southern California)
  - **Research Material Documentation**: Collected and organized research documents and papers for literature review; managed references and maintained clear documentation to support the project workflow.

- **Data Collection Support**: Gathered causal narratives from surveys, interviews, and online platforms; logged sources and key metadata to ensure traceability and research readiness.
- **Feature Preparation**: Transformed raw narratives into structured datasets by deriving interpretable indicators; documented workflows and produced summaries to support analysis and manuscript development.

- **Department of Artificial Intelligence Convergence, Chonnam National University** **South Korea**
  Research Assistant – BIEL Lab, Chonnam National University Hwasun Hospital *2021–2023*
  Research Portfolio *(Website, Slides, PDF)*
  - **AI-Driven Medical Imaging**: Leveraged deep learning models and computer vision techniques to detect and localize tumors in critical anatomical regions from medical imaging data.
  - **Research Projects**: *1. Tumor Detection in Liver Imaging:* Investigated CT, MRI, ^{18}F-FDG PET, and ^{11}C-ACT PET for liver tumor detection and segmentation.

    2. Tumor Detection in Bone Imaging: Applied thresholding, region growing, U-Net, and Mask R-CNN to bone lesion and metastatic tumor detection.

    3. Multimodal Tumor Characterization: Integrated PET, CT, and MRI information for anatomical and metabolic characterization.

    4. Optical Coherence Tomography: Combined speckle variance techniques and GANs to monitor biological tissue coagulation.
  - **Collaboration with Healthcare Professionals**: Worked with physicians and research teams to understand anatomical knowledge and clinical challenges in medical imaging.
  - **Literature Review**: Reviewed deep learning applications in medical imaging and biomedical research.

- **PropellerAds** **Singapore**
  Machine Learning Engineering *2024–Present*

- **Fintech Company** **Vietnam**
  Machine Learning Engineering and Data Mining
  - **Developing Credit Scoring Models**: Utilized regression analysis and machine learning algorithms for credit scoring, risk assessment, and financial forecasting.
  - **Risk Reduction in Credit Portfolio**: Optimized credit scoring models through feature selection and predictive modeling to support credit risk management.

Education and Courses

- **Chonnam National University – Department of AI Convergence** **Gwangju, South Korea**
  Master of Science *2021–2023*

- **Banking Academy of Vietnam – Faculty of Finance** **Hanoi, Vietnam**
  Bachelor of Arts

- **Johns Hopkins University – Department of Mathematics**
  Real Analysis (MATH 110.405) *Synchronous course*

- **CASED Center**
  Optimization *Offline course*

Scholarships and Awards

- **Scholarship Awardee: Illinois Institute of Technology**
  M.S. Applied Mathematics (Admitted with Merit Scholarship) [Proof] *Fall 2026*
  - **Graduate Excellence**: Selected for admission based on quantitative, computational, and interdisciplinary research achievements.
  - **Merit Recognition**: Awarded merit-based funding for academic excellence and research potential.

Conferences Attended

- **Oral Presentations**
  - Annual Meeting of ARCCNM-KSNM: International Conference. V.T. Nguyen, "Coagulation observation monitoring using deep learning speckle variance optical coherence tomography", South Korea.

- **Poster Presentations**

- The Korean Society of Medical & Biological Engineering. V.T. Nguyen, "Feasibility study to utilize GAN deep learning in blood coagulation monitoring with speckle variance optical coherence tomography", South Korea.

Skills

- **Programming Languages**: Python, LaTeX, R, MATLAB, HTML, CSS, C++.
- **Tools and Frameworks**: SQL, Airflow, PySpark, AWS Lambda, PyTorch, Git, Hugging Face.
- **Software**: 3D Slicer, SolidWorks, ITK-SNAP, MONAI.
- **Sequence Alignment**: BLAST, Bowtie2, Minimap2.
- **Transcriptomics Tools (RNA-Seq, scRNA-Seq)**: DESeq2, Monocle, miRDeep, miRBase.
- **Bioinformatics Pipelines**: Nextflow, Genomic Diversity Analysis, Variant Calling, Population Structure Analysis, Inbreeding Monitoring.
- **Statistical Methodology**: Multivariate Delta Method, Taylor Expansion, Asymptotic Variance, Covariance Estimation, Sample Size Calculation, Statistical Power Analysis, Directional Estimation.
- **Medical AI**: Swin UNETR, Bayesian Deep Ensembles, Conformal Risk Control, Mondrian and Weighted Conformal Prediction, Diffeomorphic Registration, Unbalanced Optimal Transport, Neural Dose Surrogates, Differentiable DVH Loss.